

MINI REVIEW OPEN ACCESS

Drug–Drug Interaction Risk Assessment Strategies for Biologics in Inflammatory Bowel Disease: A Literature-Based Evidence Mini-Review

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ABSTRACT

Biologic therapies are traditionally regarded as having low potential for drug–drug interactions (DDIs) due to their large molecular size and limited direct involvement with drug-metabolizing enzymes and transporters (DMETs). However, accumulating evidence indicates that these drug products may indirectly modulate DMET activity through cytokine-mediated mechanisms, particularly in the context of inflammatory diseases. Pro-inflammatory cytokines such as interleukin-6 have been shown to suppress the expression of cytochrome P450 (CYP) enzymes in vitro, most notably CYP3A4, thereby potentially altering the metabolic clearance of concomitant medications. This phenomenon may be pertinent in conditions such as inflammatory bowel disease (IBD) where chronic inflammation and systemic physiological alterations may significantly influence drug disposition. Although the U.S. Food and Drug Administration provides a decision framework to assess large molecule DDI risk based on the mechanism of action, it does not offer a standardized, mechanistic approach for evaluating such interactions. In response, alternative methodologies such as the use of endogenous biomarkers like 4 β -hydroxycholesterol, a proposed surrogate for CYP3A activity, and pharmacokinetic modeling are in exploration to support DDI risk assessment to potentially avoid dedicated clinical trials in patients. This review evaluates current knowledge on biologics-mediated DDIs with a particular focus on drug-disease interactions in IBD. There is an opportunity for a more structured, mechanistically informed evaluation that integrates cytokine profiling and biomarker-based approaches to identify DDI risk and enhance the overall understanding of the degree of interaction and its clinical significance.

1 | Introduction

Drug–drug interactions (DDIs) refer to the interference of one drug with at least one other resulting in altered pharmacokinetics (PK) and/or pharmacodynamics (PD) of at least one of the drugs involved. Traditionally, DDIs are linked to small molecule drugs through mechanisms involving drug-metabolizing enzymes and transporters (DMETs). However, there is increasing recognition of the importance of both drug-disease and DDIs involving biologics. These drugs refer to proteins developed as biological products for drug use, such as purified monoclonal antibodies and modified cytokines [1]. Emerging evidence suggests that biologics can indirectly influence DMET activity

through alteration of cytokine expression resulting in a DDI or drug-disease interaction.

The administration of certain biologics can alter systemic cytokine profiles by either inducing or suppressing inflammatory responses. Multiple in vitro reports show that elevated concentrations of cytokines may impact the expression and/or function of DMETs [2, 3]. Understanding how the preclinical evaluation of DDI potential of these cytokines translates to the clinic is limited perhaps with the exception of interleukin-6 (IL-6) and cytochrome P450 3A (CYP3A). To improve overall understanding and evaluate DDI potential of cytokine-modulating biologics, it is essential to adopt a systematic approach which should include

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in vitro cytokine profiling and generation of clinical data. While other small molecules such as JAK and TYK inhibitors, methotrexate, and cyclosporine decrease inflammation through cytokine modulation or through cytokine-independent pathways, these small molecules may also interact directly with DMETs which confounds the interpretation of DDI versus drug-disease interactions.

Inflammatory bowel disease (IBD) is a group of chronic autoimmune-mediated conditions characterized by increased inflammation in the gut. There are also significant disease-related consequences in other organ systems due to increased release of cytokines secondary to IBD-related inflammation. With increased systemic cytokine concentrations, there is potential for modulation of DMETs. It is important to understand how the physiological changes relate to IBD and that overall disease progression over time may impact drug exposure. This is especially relevant given that the anti-inflammatory drugs administered to suppress disease activity will change the cytokine profile and, therefore, potentially alter drug clearance through alterations in function and/or expression of DMETs.

The U.S. Food and Drug Administration (FDA) issued a guidance for DDI assessment of therapeutic proteins which includes biologics [4]. The proposed decision tree to evaluate DDI potential is based on whether the mechanism of action (MOA) is pro-inflammatory or is considered a modulator of pro-inflammatory cytokine expression. The decision tree offers the Sponsor the option to include a generic DDI warning toward CYP substrates in the FDA-approved labeling or to conduct a DDI assessment, but there are no recommendations regarding what kind of scientific assessments should be taken to comprehensively evaluate risk. The guidance document lacks a recommended systematic approach for how Sponsors should conduct DDI assessment, highlighting a gap in current understanding of how to best identify drug-disease interaction potential.

Given the significant resources required for dedicated DDI clinical studies and intractability to conduct these assessments in IBD patients, alternative strategies are often proposed to evaluate the drug-disease interaction potential of investigational biologics. Emerging evidence supports the use of endogenous biomarkers to inform DDI risk assessment [5]. Biomarker-based evaluation helps to alleviate additional risk of administering a substrate drug in a DDI trial. One such biomarker, 4 β -hydroxycholesterol (4 β OHC), a metabolite of cholesterol formed via CYP3A, has substantial evidence as a marker for CYP3A-mediated metabolic activity [6]. Altogether, biomarker evaluation approaches may identify changes in CYP metabolizing activity and can support overall drug-disease interaction risk assessment for biologics.

Another approach to evaluate DDI and/or drug-disease interaction potential is through physiologically based pharmacokinetic (PBPK) modeling. PBPK mechanistically describes drug absorption, distribution, metabolism, and elimination of drugs throughout the body and can incorporate disease-specific attributes into the model framework. Because dedicated clinical studies in IBD patients are difficult to conduct, utilizing PBPK modeling could prove advantageous to determine drug-disease interaction potential in IBD. However, given the limited number

of DDI studies conducted in IBD populations, validation of these models may be challenging.

This review explores existing evidence of drug-disease interactions in the context of IBD by examining how disease-related physiological changes may impact drug exposure. Additionally, it highlights the role of biomarker evaluation and PBPK in investigating the DDI potential of liable CYP3A substrates. By conducting a narrative literature review and analyzing available data, this work aims to summarize the current understanding of the drug-disease interaction risk and explore alternative approaches to dedicated clinical studies for drug-disease risk assessment in IBD.

2 | Evidence of CYP3A Activity Changes Secondary to Inflammation

Most evidence of cytokine-related impact is based on in vitro reports that suggest a correlation between elevated cytokine concentrations and altered DMET expression and/or function [2, 3]. These studies demonstrate an impact on multiple CYPs, but CYP3A is particularly sensitive to cytokine-induced activity changes. It is difficult to conduct in vitro to in vivo extrapolation based on in vitro data alone given that assays are not consistently conducted with cytokine concentrations at levels that are pathophysiologically relevant. While other CYPs are also impacted by cytokine modulation, this review will focus on the relationship between CYP3A and IL-6 due to the breadth of evidence available.

Studies have demonstrated change in CYP activity through a two- to five-fold change in exposure of CYP probe substrates in acute disease states [7]. Following administration of a Geneva cocktail, which included the sensitive CYP3A substrate midazolam in patients following hip surgery, a 61% reduction in CYP3A activity was observed [8], and in patients infected with SARS-CoV-2, a 22.8% reduction in CYP3A activity was reported [9]. These studies provided clinical evidence of CYP-related changes secondary to acute inflammation and point to potential DDI liability for sensitive CYP3A substrates, but questions regarding the impact of chronic inflammation on CYP substrates remain.

While in vitro studies report that multiple cytokines may down-regulate CYP3A, IL-6 consistently emerges as the most potent contributor [2, 3, 10–12]. Thus, in immune-mediated inflammatory diseases like IBD that are characterized by systemically elevated cytokines, CYP3A activity is potentially decreased compared to healthy subjects. Most IBD treatments decrease pro-inflammatory cytokine levels, which may lead to normalization of CYP3A levels after dosing and potentially increased clearance of liable CYP3A substrates. It is not established what IL-6 concentration translates to a clinically meaningful DDI risk. IL-6 changes are highly variable within patients, between diseases, and over time. This limits the generalizability of IL-6 levels to evaluate DDI risk across different inflammatory diseases. Multiple cytokines that demonstrated in vitro potential to modulate DMETs are elevated alongside IL-6, which makes it challenging to isolate the role of IL-6 from other cytokines. Moreover, the bioanalytical assays used to quantify IL-6 have variable performance and may limit between-study

comparisons. These findings raise questions regarding the utility of IL-6 as an indicator of DDI risk and if a specified IL-6 threshold and/or reduction could be widely applicable in systematic DDI assessment.

While limitations of assay sensitivity and in vitro to in vivo correlation make it challenging to quantify a specific IL-6 threshold for disease-DDI risk, dedicated clinical DDI studies have demonstrated a relationship between direct IL-6 modulation and CYP3A activity in rheumatoid arthritis (RA) patients. A decrease in overall exposure was observed in simvastatin and midazolam (sensitive CYP3A substrates) following administration of monoclonal antibodies that targeted the IL-6 pathway [13–15]. Simvastatin exposure was reduced by 30%–60% whereas midazolam was reduced by 30%–50%. Although the overall DDI magnitude is considered moderate (two to five-fold overall change in exposure) and transient, these results support the translation of in vitro CYP3A activity changes to a potentially relevant in vivo change in drug exposure, particularly for narrow therapeutic index drugs.

However, it is not established if the results of these studies can be extrapolated to drugs with different mechanisms of action and/or to other inflammatory diseases (i.e., from RA to IBD). One clinical study evaluated the impact of risankizumab (IL-23 binding monoclonal antibody) administration on multiple CYP probe substrates to characterize drug-disease interactions in moderate to severe IBD patients [16]. In contrast to the studies of IL-6-targeting biologics, there was no clinically meaningful impact on CYP3A nor on any of the other CYPs tested in this study. Taken together, the data from these studies indicate the importance of interpreting clinical results in the specific context of both disease and MOA and caution against extrapolation to other patient populations or drugs that exert an immunomodulatory effect by acting on other parts of the cytokine cascade.

3 | IBD-Related Impacts on Cytokine Levels and Protein Expression

IBD is a group of chronic immune-mediated inflammatory conditions that deal with physiological changes secondary to

inflammation that primarily manifest within the GI tract. This subset of diseases primarily encompasses two conditions: ulcerative colitis (UC) and Crohn's Disease (CD). While the two conditions differ in their pathological manifestations, they share overlapping etiological features and are managed with similar therapeutics. Given the disease-mediated changes of cytokines, there is potential for modulation of DMETs both systemically and at the gut level at baseline and after administration of anti-inflammatory therapy.

IBD is associated with elevated concentrations of several pro-inflammatory cytokines, and as demonstrated, IL-6 stands as the main perpetrator of interest toward CYP3A at the gut and hepatic level. Table 1 highlights an increase in systemic IL-6 concentrations at baseline in UC and CD patients versus healthy controls. There is no trend in Table 1 that suggests a difference in systemic IL-6 between UC and CD which is important given that CD is considered a more severe inflammatory state and further suggests the IL-6-related drug–disease interaction risk is not different between UC and CD.

Understanding changes in CYP3A at the gut level is important given the locality of disease manifestations of IBD and high level of CYP3A expression throughout the GI tract. There are no reports of local cytokine concentrations in IBD patients, which limits understanding of cytokine-related DMET changes at the gut level. Through biopsy of different segments of the intestine, CYP3A protein content was reduced in the ileum and ascending colon by 45% and 78%, respectively, in CD patients [18]. It is not clear, however, how the impact of the CYP3A downregulation separates from disease-related physiological changes, such as villi length reduction, on the overall impact on drug bioavailability.

IBD also affects the expression of drug-binding proteins. During active disease, levels of albumin and bilirubin (total, direct, and indirect) are typically reduced while concentrations of alpha-1-acid glycoprotein (AAG) are elevated [19, 20]. The opposing trends in albumin and AAG levels underscore the need for drug-specific considerations when evaluating the IBD impact on pharmacokinetics, especially in distinguishing between total and unbound drug concentrations. This could play a confounding

TABLE 1 | Comparison of reported levels of baseline IL-6 concentrations in UC and CD to healthy subjects (modified from Sun et al., *Clin Pharmacol Drug Dev*, 2021) [17].

IL-6 levels in IBD patients vs. healthy subjects (pg/mL)			
Study citation	UC	CD	Healthy subjects
Holtkamp et al.	10 ± 4 ^a	36 ± 8 ^a	7.3 ± 1.2
Szkaradkiewicz et al.	8.63 ± 2.14 ^a	8.24 ± 1.75 ^a	1.59 ± 0.9
Martinez-Fierro et al.	14.4 ± 3.4	18.1 ± 1–0.6	14.4 ± 10.7
Korolkova et al. (median/IQR)	0 (0–1.49) ^a	1.53 (0–4.85) ^a	0 (0–0.97)
Ciecko-Michalska et al. (median/IQR)	19.6 (21) ^a	10.8 (7.6) ^a	3.2 (1.6)
Biesiada et al.	8.03 ± 0.7 ^a	n/a	5.13 ± 0.40

Note: Table values present average ± standard error unless noted otherwise. Abbreviations: CD = Crohn's Disease, n/a = not collected, UC = ulcerative colitis. ^aDenotes statistical significance between diseased versus healthy values.

role when eliciting mechanisms behind CYP3A-related drug-disease interactions for concomitantly administered drugs with a high degree of plasma protein binding.

Anti-inflammatory treatments are integral in IBD management and aim to decrease pro-inflammatory cytokine levels, including IL-6, thereby implying that sensitive CYP3A substrates may have decreased exposure after administration of immunomodulating therapies. However, the presented evidence demonstrates the difficulty of eliciting CYP3A-mediated impact on drug exposure versus other disease-related physiological changes.

4 | Selected Reports of Drugs With CYP3A Clearance With Altered Exposure in IBD

Given the well-documented cytokine elevation and physiological alterations in IBD, it is important to understand the impact on PK and PD of concomitant therapies. The greatest risk for drug-disease interactions would follow the first dose of anti-inflammatory treatment or during a treatment flare up, as both cases would indicate significant alterations in cytokine concentrations. This is especially relevant as most IBD patients are often polymedicated to manage their condition, and it is important to understand the DDI potential of co-administered drugs so as not to compromise safety and efficacy in these patient populations.

To determine how drug-disease interactions manifest in CD, one review examined exposure changes in small molecule drugs in patients [21]. As CD generally presents more severe systemic inflammation than UC, the findings from this review may be classified as a worst case scenario for IBD patients. Among the CYP3A substrates evaluated (Table 2), the reported changes in exposure were variable in magnitude and direction. This variability highlights the heterogeneity of IBD and underscores the complexity of characterizing drug exposure changes in this population.

A particularly noteworthy case example was the five-fold difference in midazolam clearance reported between healthy controls

and patients, suggesting significantly reduced CYP3A activity. Midazolam is recommended by the FDA as a clinical substrate for evaluating CYP3A-related DDI potential, and results of CYP3A DDI assessments with midazolam can be extrapolated to other drugs metabolized by CYP3A. However, several factors limit the interpretability of these results. All patients received budesonide therapy for at least 1 month prior to receiving midazolam and were classified as in remission based on the Harvey-Bradshaw Index score. Therefore, the applicability of these results to patients with active disease remains limited, especially in those who are treatment naive. There are also inconsistencies in the reported PK parameters for midazolam, specifically the direction of change in bioavailability and clearance, which pose questions about potential analytical issues in the study.

Verapamil was also highlighted in this report with a substantial change in exposure [23]. The reported change in the (R)- and (S)-verapamil enantiomers plasma concentrations is significantly increased as shown by the total drug levels in patients with active disease versus those in remission. However, the increase in PK did not translate into enhanced PD since no corresponding changes were observed in blood pressure, heart rate, or PR interval prolongation. The lack of concordance between elevated total drug concentrations and unchanged PD outcomes raises the possibility of altered protein binding in active disease. Since IBD is associated with alterations in drug-binding proteins such as decreased albumin and increased AAG [19, 20], this may result in a disproportionate increase in total versus unbound (i.e., pharmacologically active) drug concentrations. Furthermore, verapamil is a dual substrate of CYP3A and P-glycoprotein (P-gp) which is important due to potential P-gp downregulation in inflammatory conditions [24]. The reduction in P-gp expression could contribute to increased bioavailability through decreased intestinal efflux, further complicating the overall interpretation of exposure–response relationships and DDI magnitude.

The final example from this review illustrating the complexity of the drug-disease phenomenon in IBD are the inconsistencies

TABLE 2 | Selected reports of CYP3A substrates with reported exposure changes in CD patients [21]. Elimination and disposition pathways and total clearance values collected from Certara Drug Interaction Database [22].

Selected reports of drugs with altered exposure in CD patients			
Drug	Elimination and disposition pathway(s)	Total clearance (L/min)	Observed change in CD patients versus healthy subjects
Midazolam	CYP3A ($f_m = 0.96$), $f_e < 0.005$	0.29–0.63	↑ 5-fold exposure ↓ 5-fold clearance
Alfentanil	CYP3A ($f_m = 0.97$), $f_e = 0.01$	0.23	↔ clearance
Budesonide	CYP3A ($f_m = 0.76$), P-gp	1.0	Mixed reports
Cyclosporine	CYP3A ($f_m = 0.79$), P-gp, $f_e = 0.01$	0.3	↔ exposure
Verapamil	CYP2C8, CYP3A, P-gp	R—0.66. S—2.86	↑ total exposure (~10-fold ↑ [S]-verapamil, ~2-fold ↑ [R]-verapamil). ↔ unbound exposure
Prednisolone	$f_e = 0.98$, CYP3A	0.14	Mixed reports

Abbreviations: CD = Crohn's Disease, CYP = cytochrome P450, f_e = fraction excreted in urine, f_m = fraction of clearance through metabolism by a given CYP, P-gp = P-glycoprotein, UC = ulcerative colitis.

in budesonide exposure changes [21]. Budesonide is one of the early treatment options for CD and has the most evidence of exposure in CD patients in comparison to healthy volunteers. As a substrate for both CYP3A and P-gp, increased systemic exposure is expected in the setting of downregulated metabolic and efflux pathways. However, this review highlighted variable case reports of budesonide exposure changes in CD patients, notably cases with increased, decreased, or no change in exposure in both fed and fasted states.

Altogether, these examples underscore the multifactorial nature of drug disposition in IBD. While cytokine-mediated downregulation of DMET is well-documented in vitro, the heterogeneity of IBD and physiological alterations can lead to unpredictable PK changes. These findings caution against assuming uniform alterations in drug exposure across the IBD population, and furthermore, the mixed evidence suggests that multiple confounding factors may play a role in exposure beyond cytokine-mediated CYP changes.

5 | Use of 4βOHC as a CYP3A Biomarker to Determine DDI Effects

Assessing drug-disease interactions in the context of IBD and drug development remains a significant challenge in drug development. Due to the limited ability of sponsors to enroll patients into a dedicated DDI clinical trial, alternative investigations are proposed to evaluate DDI liability after drug administration. This review focuses on the potential utility of the cholesterol metabolite 4β-hydroxycholesterol (4βOHC) as a biomarker for systemic CYP3A activity.

Cholesterol is primarily metabolized in the liver via CYP7A1 and CYP27A1 into downstream metabolites and specifically by CYP3A into 4βOHC [6]. Limited information about the transport of 4βOHC is known, but its metabolism primarily follows the same CYP pathways as cholesterol (i.e., CYP7A1, CYP27A1) [25]. It is unclear how much intestinal CYP3A contributes to the generation of 4βOHC or how much of a role CYP3A5 polymorphisms may play in overall disposition. Consequently, the roles of transport and genetic variability in CYP3A5 in modulating 4βOHC levels before and after administration of a perpetrator drug are not fully understood.

Multiple reports have assessed the changes in 4βOHC secondary to CYP3A inhibitors and inducers [26, 27]. Overall, 4βOHC demonstrates greater sensitivity to CYP3A induction than CYP3A inhibition given that the cholesterol and 4βOHC half-lives are relatively long (46 and 17 days, respectively). Therefore, this biomarker is potentially well suited to monitor disease changes before and after treatment since disease modification is a process that takes a significant amount of time. Given its sensitivity to induction, 4βOHC may aid in understanding the changes in CYP3A activity observed before and after administration of anti-inflammatory therapies. While the magnitude of predicted drug-disease interaction may not achieve the degree of a true induction DDI per se, longitudinal changes in the 4βOHC levels in the stages of drug administration from baseline through disease maintenance could provide insights into CYP3A activity modulation and if there is a significant change that can be detected within the IBD population.

A case study involving vedolizumab illustrates this application in the context of IBD (Figure 1) [17]. With healthy subject controls to serve as a comparator, IL-6, IL-10, IL-8, IFN-γ, and TNF-α changes were reported in CD patients at baseline prior to initiation of treatment following induction of vedolizumab and during maintenance dosing (Figure 1A–E). Although modest changes in the cytokine levels were observed in CD patients after treatment was administered, no significant change in the 4βOHC ratio to cholesterol was detected (Figure 1F). These findings support the lack of a clinically meaningful disease-drug interaction involving CYP3A following treatment with vedolizumab.

Although 4βOHC as an evaluative biomarker is not without its caveats, it remains a valuable tool for evaluating CYP3A activity in IBD and supporting decision-making against the presence of DDI potential in the development of a drug that alters inflammatory pathways.

6 | The Role of Physiologically-Based Pharmacokinetic Modeling in Biologic-DDI Assessment

Physiologically based pharmacokinetic (PBPK) modeling is a modeling and simulation approach used to mechanistically

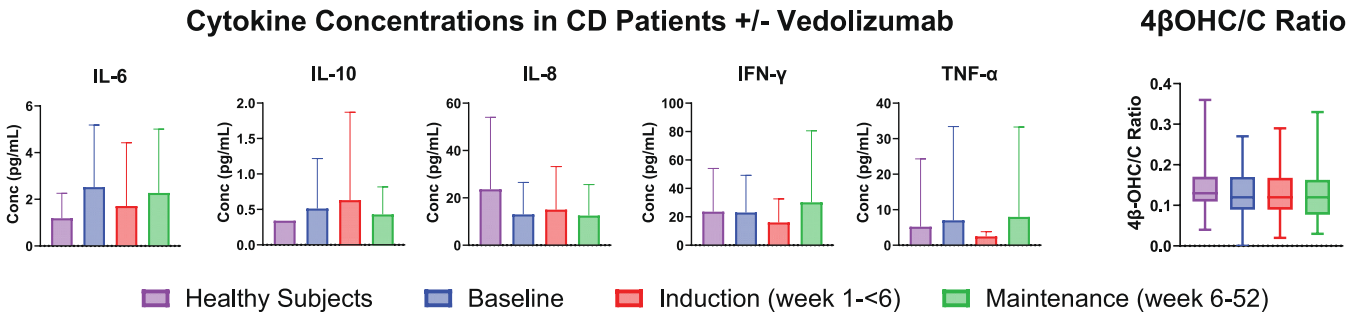


FIGURE 1 | Adapted from Sun et al., *Clin Pharmacol Drug Dev*, 2020 [17]. Findings of cytokine and 4βOHC/C ratio assessment in CD patients in various stages of treatment vs. healthy subjects. Concentrations of cytokines in CD patients before and after vedolizumab therapy compared to healthy volunteer data are shown for IL-6 (A), IL-10 (B), IL-8 (C), IFN-γ (D), and TNF-α (E). To demonstrate lack of change in CYP3A activity, 4β-hydroxycholesterol (4βOHC) was taken in ratio to cholesterol (C) in CD vs. healthy subjects (F).

understand drug elimination and its distribution in the body through multiple compartments designed to anatomically represent organ systems and physiological processes. Given that one of the most effective applications of PBPK is to evaluate DDI risk of new drugs, this highlights an opportunity to use this technique to evaluate drug–disease interaction potential.

Multiple publications demonstrated that PBPK models can characterize drug-disease interactions while primarily focusing on the role of IL-6 on CYP suppression. One review aimed to determine if IL-6 alone as the primary precipitant can predict the magnitude of the drug-disease interaction [28]. The simulations captured the observed drug-disease interactions from the clinical studies conducted in RA patients described earlier within a reasonable margin of error. It should be pointed out that other cytokines were not included as precipitants even though in vitro evidence suggests that they may contribute to altered DMETs. The highlighted PBPK models only validated model performance against observed clinical studies in RA patients, which may limit applicability to other disease states like IBD due to lack of observed DDI study data in IBD patients [29–31]. Validation of PBPK models for evaluating drug-disease potential in IBD would be difficult to conduct for the same reason—there are no studies to ensure the model reliably predicts the interaction risk. Another recent publication characterized the current state of IL-6-based PBPK models and the predictive power for drug-disease interactions [32]. Of particular note, an issue that is consistent across models is the lack of complexity in the IL-6 mechanisms related to CYP suppression. Most existing models also assume uniform effects in the liver and gut and do not account for differences in the IL-6 concentrations nor IL-6 receptor dynamics at a tissue-specific level. In summary, there are still considerable gaps that need to be addressed prior to acceptance of PBPK models employing IL-6 as the primary precipitant for characterizing drug-disease potential in IBD.

Furthermore, appropriate disease-related GI physiological changes should be applied appropriately in a PBPK model format. Some examples of physiological changes specific to IBD that may impact drug exposure include altered DMET expression and function, altered plasma protein binding, cytokine concentrations in the gut lumen and GI tissue, changes in gut motility, reduction in microvilli surface area, and altered gut wall permeability [33]. There are differences in disease presentation of UC and CD, which further raises questions regarding application of PBPK models interchangeably between these two disease states.

A key consideration when assessing the complex role of physiological factors at the gut level for model parameterization is the considerable heterogeneity in disease presentation. This may limit parameter and assumption validations given that the direct impact on drug exposure from specific pathophysiologic changes may not be mutually exclusive. Additionally, existing models show poor predictive performance when protein binding is altered as a consequence of disease [34], and case examples of sensitive CYP3A substrates in IBD, as highlighted in this review, demonstrate conflicting results in the direction of exposure change. These examples further support the idea of non-mutually exclusive, confounding disease factors on

co-administered drug exposure which may complicate PBPK model development.

While PBPK presents opportunities to support evaluation of drug-disease interactions, the current applications of this tool may be limited in IBD given the lack of consensus on the role of IL-6 and other cytokines on DMETs and the role of disease-specific physiological changes that may impact drug exposure.

7 | Conclusions and Remaining Gaps in Understanding

While emerging evidence has shed some light on drug-disease interactions, there are still gaps in current understanding regarding how to best identify drug-disease DDI potential in IBD. There is no definitive evidence of consistent CYP3A downregulation at baseline, nor is there clarity on the degree of increase in CYP3A activity after re-normalization of protein levels. Thus, it is unknown if these changes translate to clinically meaningful alterations of drug exposure following initiation of anti-inflammatory therapy in IBD.

It is not clear if existing clinical evidence of drug-disease interactions shown in certain immune-mediated diseases translates to IBD. Dedicated studies demonstrated a clear CYP3A effect in RA patients after administration of drugs that directly target IL-6. The most effective and commonly used biologic therapies in IBD do not directly modulate IL-6; rather, they work through modulation of other cytokines to decrease overall pro-inflammatory cytokine expression which will indirectly impact IL-6 levels. The example of risankizumab, an anti-IL-23 monoclonal antibody which did not demonstrate a clinically meaningful effect on CYPs in IBD patients [16], emphasizes that the RA observations may not translate.

Even though some evidence suggests altered CYP3A activity in inflammatory states, the data currently available do not provide definitive conclusions on drug-disease interaction risk, and multiple confounding variables related to disease and resulting inherent variability of drug exposure may impede interpretability of results. It is unlikely that current in vitro assays evaluating cytokine-related drug-disease interaction potential are sufficient to fully characterize the drug-disease potential based on the evidence presented in the review. Based on this rationale, use of PBPK models in IBD to predict drug-disease interactions may also be limited in the foreseeable future.

4 β OHC has shown promise as a biomarker for assessing overall systemic changes in CYP3A activity and may help support drug-disease interaction assessment in an IBD population, but likely does not address the changes in CYP3A at the gut level. Caveats exist in the use of 4 β OHC, and given that the degree of CYP3A re-normalization after treatment evaluation is likely not to the magnitude of induction-based DDIs, questions remain regarding the sensitivity of 4 β OHC for such assessments.

A combined, mechanistically informed evaluation that integrates cytokine profiling and biomarker-based approaches may ultimately be needed to assess overall drug-disease risk and

enhance the overall understanding of the degree of interaction and its clinical significance in IBD.

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Conflicts of Interest

C.S., S.E.S., T.B., S.A.-O. are employees of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co. Inc., Rahway, NJ, USA and may hold stock options in Merck & Co. Inc., Rahway, NJ, USA.

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